



Overview

Personal Weather Stations (PWS) Historical Data returns the historical PWS data for a single date, returning hourly data, summary data for the entire day, or both.

The primary use case for this data is to view historical data from previous dates. Use cases that require past data from the current day should leverage alternative PWS APIs.

HTTP Headers and Data Lifetime - Caching and Expiration

For details on appropriate header values as well as caching and expiration definitions, please see [The Weather Company Data | API Common Usage Guide](#).

Unit of Measure Requirement

The unit of measure for the response. The following values are supported:

- e = Imperial (English) units
- m = Metric units

Atomic Endpoints	Aggregate Product Name
v2/pws/history/hourly	v2-pws-history-hourly
v2/pws/history/all	v2-pws-history-all
v2/pws/history/daily	v2-pws-history-daily

URL Construction

Atomic API URL Examples:
Request by PWS Station ID: Required Parameters: <code>stationId</code> , <code>format</code> , <code>units</code> , <code>date</code> , <code>apiKey</code> Optional Parameter: <code>numericPrecision</code> , <code>startDate</code> , <code>endDate</code> <code>https://api.weather.com/v2/pws/history/hourly?stationId=<stationId>&format=<format>&units=<units>&date=<date>&apiKey=<apiKey></code>
<code>https://api.weather.com/v2/pws/history/hourly?stationId=KCAOAKLA44&format=json&units=m&date=20181001&apiKey=yourApiKey</code>

Valid Parameter Definitions

Parameter Name	Valid Parameter Value	Description
stationId	KCAOAKLA44	Any valid PWS Station ID. To be used as location query parameter.
date	YYYYMMDD	The date parameter is used to call the specific date request, using format “YYYYMMDD”.

startDate	YYYYMMDD	The startDate parameter is used to request a range of days, using format “YYYYMMDD.” Must be used with the endDate parameter. The startDate and endDate range is limited to 31 total days. This query parameter is limited to use on the hourly and daily endpoints.
endDate	YYYYMMDD	The endDate parameter is used to request a range of days, using format “YYYYMMDD.” Must be used with the startDate parameter. The startDate and endDate range is limited to 31 total days. This query parameter is limited to use on the hourly and daily endpoints.
numericPrecision	decimal	Optional parameter. Set to ‘decimal’ to ensure data is returned in decimal format when needed. Will return integers if this value is not used.

Data Elements & Definitions

Note: Field names are sorted alphabetically in the table below for presentation purposes. The table below does not represent the sort order of the API response.

Field Name	Description	Type	Range	Sample	Nulls Allowed
epoch	Time in UNIX seconds	epoch		1369252800	Y
humidityAvg	Average Humidity of the period	integer/decimal		32	Y
humidityHigh	Highest Humidity of the period	integer/decimal		32	Y
humidityLow	Lowest Humidity of the period	integer/decimal		32	Y
lat	Latitude of PWS	decimal	Any valid latitude value. -90 to 90	29.8972	Y
lon	Longitude of PWS	decimal	Any valid longitude value. -180 to 180	-97.9362	Y
obsTimeLocal	Time observation is valid in local apparent time by timezone	ISO	YYYYY-MM-dd HH:mm:ss	2016-09-27 00:59:39	Y
obsTimeUtc	GMT(UTC) time	ISO	ISO 8601 - yyyy-MM-dd'T'HH:mm:ssZZ	2016-09-27T06:59:39Z	Y
solarRadiationHigh	Highest Solar Radiation of the period	integer/decimal		947.00	Y
stationID	ID as registered by wunderground.com	string		KAZTUCSO539	N
tz	Time zone of PWS	string		America/Chicago	Y
uvHigh	Highest UV Index of the period	integer/decimal		2.00	Y
winddirAvg	Wind direction average of the period	integer/decimal		170	Y
Imperial (english) metric	Object containing fields that use a defined unit of measure. The object label is dependent on the units parameter assigned in the request. "imperial", "metric"	object		imperial: {...}	N
dewptAvg	Average dew point of the period	integer/decimal	-80 to 100 (°F) or -62 to 37 (°C)	43	Y
dewptHigh	Maximum dew point of the period	integer/decimal	-80 to 100 (°F) or -62 to 37 (°C)	43	Y
dewptLow	Minimum dew point of the period	integer/decimal	-80 to 100 (°F) or -62 to 37 (°C)	43	Y
heatindexAvg	Heat index average of the period	integer/decimal		68	Y
heatindexHigh	Heat index high temperature of the period	integer/decimal		71	Y
heatindexLow	Heat index low temperature of the period	integer/decimal		61	Y
precipRate	Rate of precipitation - instantaneous precipitation rate. How much rain would fall if the precipitation intensity did not change for one hour	integer/decimal		0	Y
precipTotal	Accumulated Rain for the day ffrom midnight to present in defined unit of measure	integer/decimal		0	Y

pressureMax	Highest Barometric pressure in defined unit of measure of the period	integer/decimal		30.12	Y
pressureMin	Lowest Barometric pressure in defined unit of measure of the period	integer/decimal		0.01	Y
pressureTrend	Pressure tendency over the preceding period	integer/decimal		28.09	Y
qcStatus	Quality control indicator: -1: No quality control check performed 0: This observation was marked as possibly incorrect by our quality control algorithm 1: This observation passed quality control checks	integer	-1 to 1	1	N
tempAvg	Temperature average of the period	integer/decimal		73	Y
tempHigh	High Temperature of the period	integer/decimal		87	Y
tempLow	Low Temperature of the period	integer/decimal		64	Y
windchillAvg	Windchill average of the period	integer/decimal		32	Y
windchillHigh	High Windchill temperature of the period	integer/decimal		45	Y
windchillLow	Low Windchill temperature of the period	integer/decimal		35	Y
windgustAvg	Wind gust average of the period	integer/decimal		54	Y
windgustHigh	Highest Wind gust of the period	integer/decimal		56	Y
windgustLow	Lowest Wind gust of the period	integer/decimal		43	Y
windspeedAvg	Wind speed average of the period	integer/decimal		3	Y
windspeedHigh	Highest Wind speed of the period	integer/decimal		5	Y
windspeedLow	Lowest Wind speed of the period	integer/decimal		1	Y

JSON Sample

```
// Response Collapsed for Presentation Purposes
{
  "stationID": "KMAHANOV10",
  "tz": "America/New_York",
  "obsTimeUtc": "2019-02-02T05:04:58Z",
  "obsTimeLocal": "2019-02-02 00:04:58",
  "epoch": 1549083898,
  "lat": 42.09263229,
  "lon": -70.86485291,
  "solarRadiationHigh": 940.00,
  "uvHigh": 2.00,
  "winddirAvg": 225,
  "humidityHigh": 73,
  "humidityLow": 73,
  "humidityAvg": 73,
  "qcStatus": 1,
  "imperial": {
    "tempHigh": 8,
    "tempLow": 7,
    "tempAvg": 7,
    "windspeedHigh": 0,
    "windspeedLow": 0,
```

```
    "windspeedAvg": 0,  
    "windgustHigh": 0,  
    "windgustLow": 0,  
    "windgustAvg": 0,  
    "dewptHigh": 0,  
    "dewptLow": 0,  
    "dewptAvg": 0,  
    "windchillHigh": 8,  
    "windchillLow": 7,  
    "windchillAvg": 7,  
    "heatindexHigh": 8,  
    "heatindexLow": 7,  
    "heatindexAvg": 7,  
    "pressureMax": 30.26,  
    "pressureMin": 30.25,  
    "pressureTrend": -0.12,  
    "precipRate": 0,  
    "precipTotal": 0  
  }  
}
```